HEALTHCARE ACCESS INTELLIGENCE

Where America runs short on doctors, and how to fix it

A plan for where to invest in primary care first, built from data on 3,220 U.S. counties

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71.5M
people live where doctors are scarce

~20,400
more doctors needed, at most

4
county types, each with its own fix

71.5M Americans live where doctors are scarce, and rural, poorer places are hit hardest

The problem

71.5M people live in places the government officially marks as short on doctors. That is about **1 in 5 Americans**.

Where it's worst

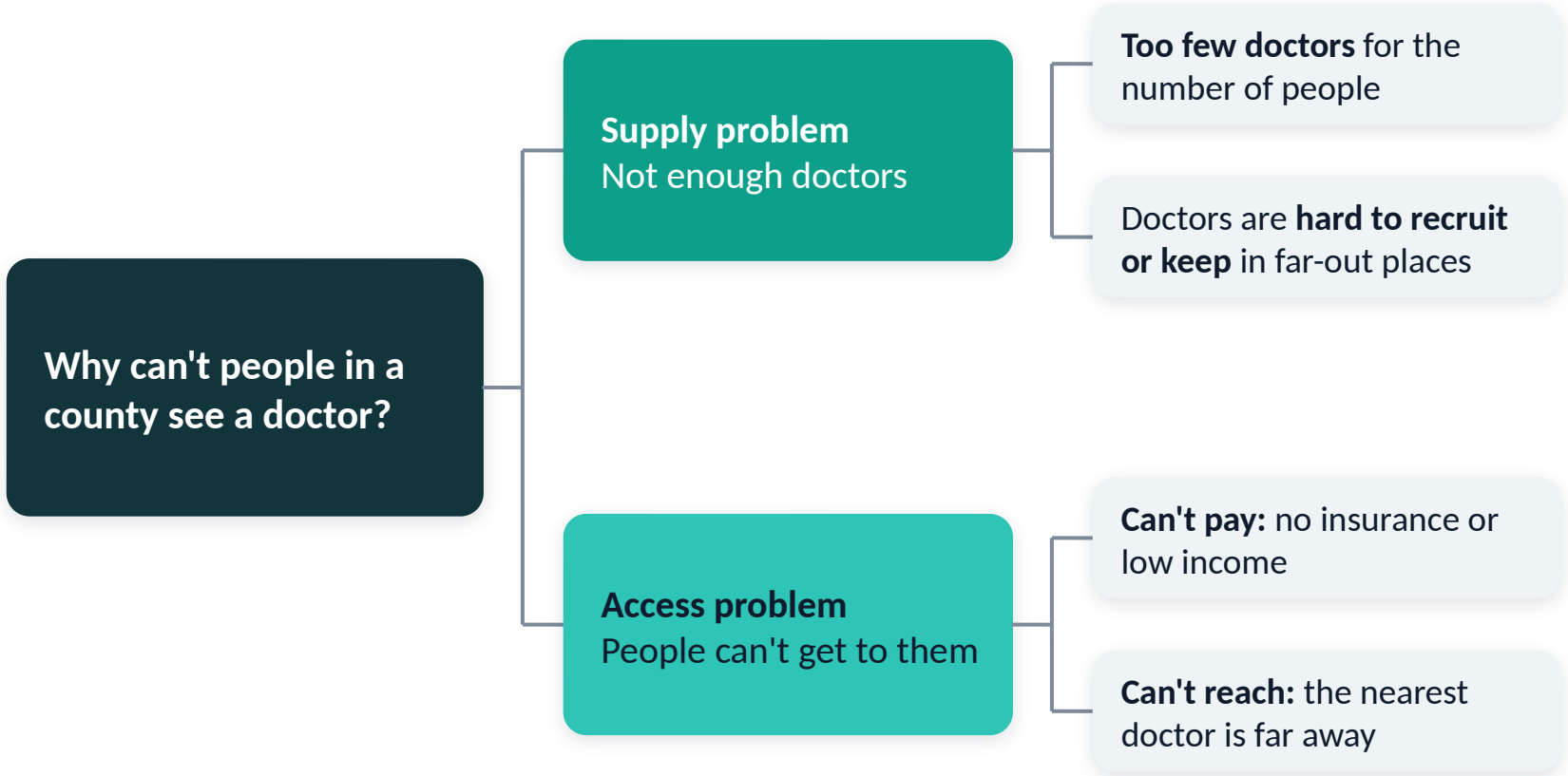
82% of the 50 worst counties are **rural** (vs 61% of all counties). They have **more uninsured** people (12.1% vs 8.5%) and **less money**.

What to do

Match the fix to the place: **sign people up for insurance**, bring care closer with **telehealth**, and add **community health workers** in cities.

Bottom line: up to **~20,400 more doctors** are needed. Start where the need is deepest: **544 rural counties** where many people have no insurance.

Two simple reasons people can't see a doctor: too few doctors, or no way to reach them



WORD TO KNOW

HPSA

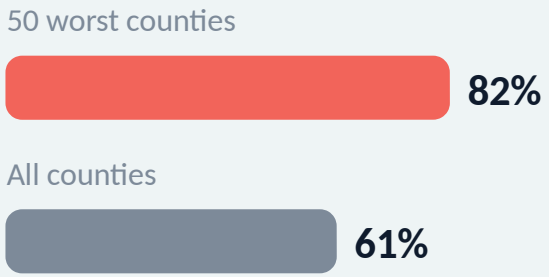
Health Professional Shortage Area: a place the government officially says has too few doctors.

HOW WE CHECKED

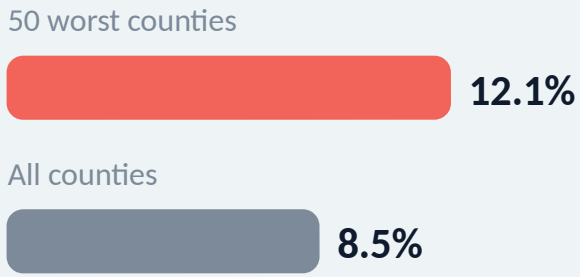
- **Census:** income, insurance
- **HRSA:** shortage areas, scores
- **USDA:** how rural (1-9)

The worst shortages land in rural places with less money and more uninsured people

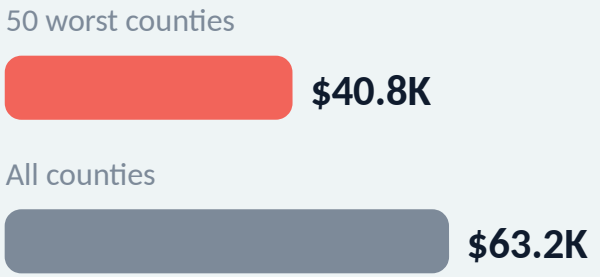
Rural counties



No health insurance



Household income



We tested three guesses

Rural places have worse shortages. Confirmed: shortage score 13.9 vs 11.8.

More uninsured, worse shortages. Confirmed, moderate link (r = 0.25 on a 0-1 scale).

Poorer places have more shortage areas. Not as tested: bigger counties just have more areas.

Not every short-on-doctors county is the same: four types need four different fixes

Rural & Uninsured

Small, far-out towns where many people have no insurance

544
counties

13.3M people
13.6% uninsured
Shortage score 17.7

Rural & Remote

Small towns where most people are insured, but the doctor is far

298
counties

7.4M people
6.0% uninsured
Shortage score 17.1

Urban Underserved

Cities and suburbs with doctors nearby, but not enough of them

449
counties

130.6M people
9.1% uninsured
Shortage score 17.3

Lower Priority

Places with milder shortages. Keep an eye on them

1,929
counties

184.4M people
8.4% uninsured
Shortage score 10.2

How we sorted: priority = shortage score above the national middle; rural = USDA code 4-9; high uninsured = above the middle county. **Shortage score** = HRSA's 0-25 rating, higher is worse. A computer clustering check (k-means) mostly agreed.

Closing the gap would take up to ~20,400 more primary care doctors

71.5M
people living in shortage areas

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3,500
people per doctor (HRSA's rule of thumb)

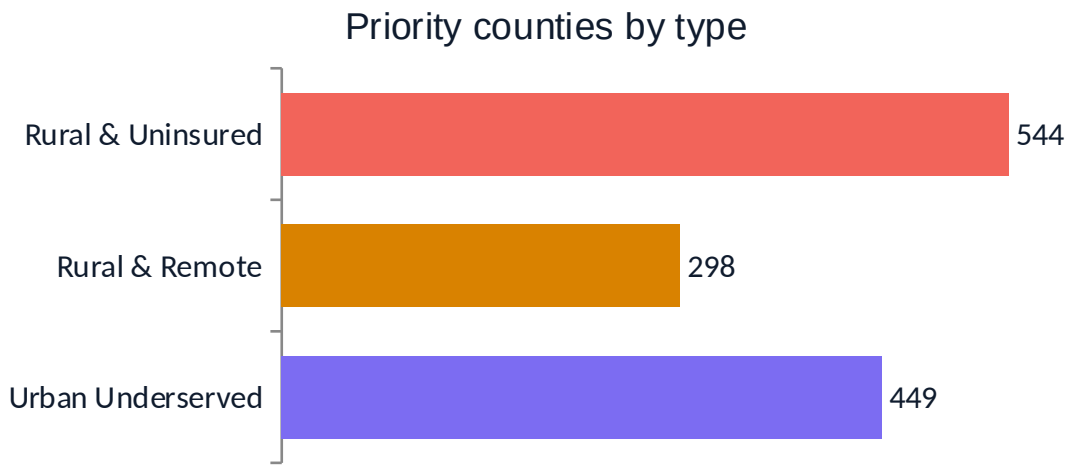
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~20,400
more doctors needed, at most

What this means

If these areas had **zero** doctors, they would need about 20,400. Some doctors already work there, so the **real gap is smaller**. Think of 20,400 as the **ceiling**.

Clinic-based shortage designations were left out so nobody is **counted twice**.



Start with three fixes, each matched to the county type it helps most

1

Rural & Uninsured | 544 counties

Sign people up for **Medicaid** and open more clinics (**FQHCs**)

Why: highest shortage score (17.7) and most uninsured (13.6%)

2

Rural & Remote | 298 counties

Bring care to them: **telehealth** and **mobile clinics**

Why: people have insurance; distance is the problem

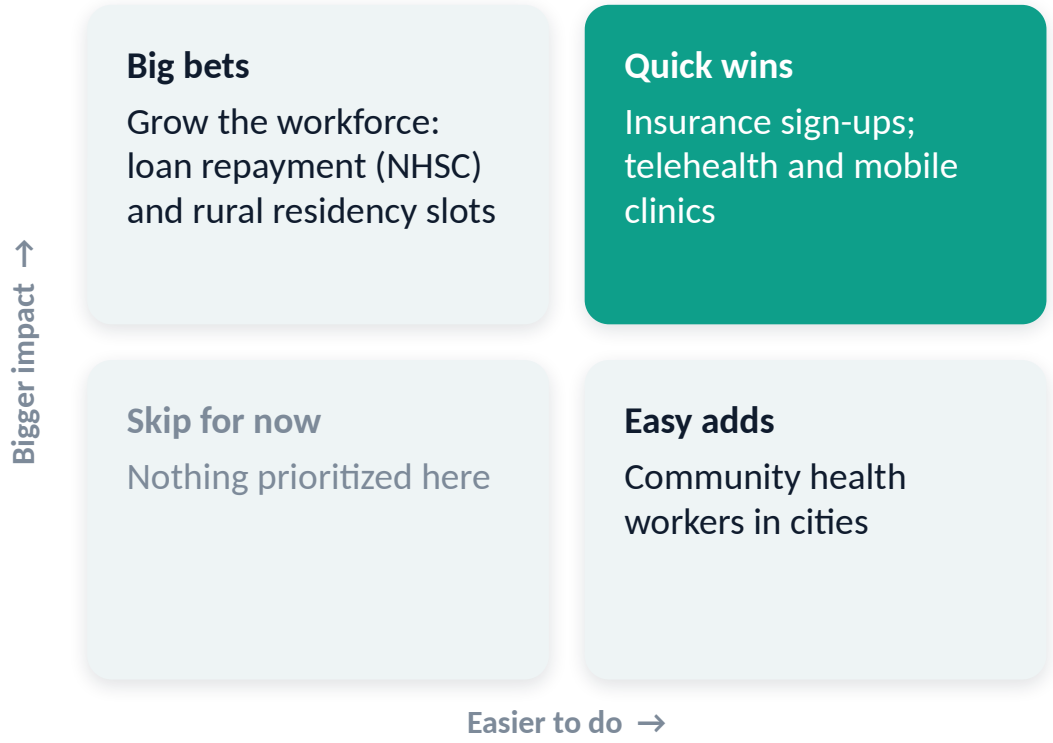
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Urban Underserved | 449 counties

Add **community health workers** and longer clinic hours

Why: reaches the most people (130.6M live in these counties)

Impact vs. effort



APPENDIX

How we know: data, model accuracy, and what this analysis can't tell us

Data we used

Census ACS (2023): people, income, and % uninsured for every county

HRSA HPSA (Oct 2026): official shortage areas and their 0-25 severity scores

USDA RUCC (2023): how rural each county is, from 1 (big city) to 9 (most rural)

How good is the model?

0.70 AUC

AUC scores how well the model tells high-risk from low-risk counties: 0.5 = coin flip, 1.0 = perfect.

Accuracy **62.5%** vs 59.9% from always guessing the most common answer.

Strongest clues: **county size** and **income**.

What this can't tell us

- County-level data: shortage areas often cover only part of a county
- The model shows links, not causes
- Shortage areas are matched by county name, so a few may be missed
- 20,400 is a ceiling; income isn't adjusted for cost of living